

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Mathematics 1

- 1 A bag contains n green counters and $(n - 2)$ red counters, where $n \geq 3$.

Two counters are drawn at random from the bag without replacement.

The probability that the two counters are of different colours is $\frac{8}{15}$.

What is the sum of all possible values of n ?

- A 4
- B 6
- C 8
- D 10
- E 12
- F 16
- G 24

- 2 A set of 200 quadratic equations is considered. Each equation has exactly two real roots, and none of the roots are zero.

It is found that:

- 30% of the equations have two positive roots.
- 45% of the equations have two negative roots.

The remaining equations each have one positive and one negative root.

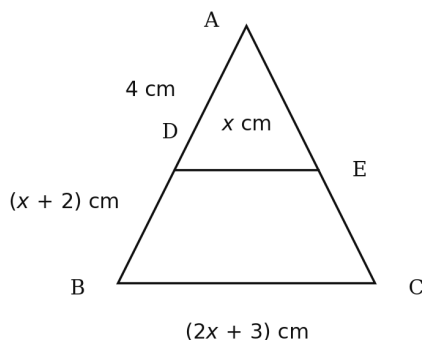
What is the total number of positive roots and negative roots, respectively, in the set?

- A (110, 140)
- B (110, 230)
- C (120, 180)
- D (170, 140)
- E (170, 230)
- F (230, 170)

- 3 In the triangle ABC shown below, points D and E lie on the sides AB and AC respectively, such that DE is parallel to BC .

$AD = 4$ cm, $DB = (x + 2)$ cm, $DE = x$ cm, and $BC = (2x + 3)$ cm.

What is the length, in cm, of BC ?



[diagram not to scale]

- A $1 + \sqrt{13}$
 B $1 + 2\sqrt{13}$
 C $3 + \sqrt{21}$
 D $5 + 2\sqrt{13}$
 E 9
 F $9 + 2\sqrt{21}$
- 4 The table shows summary statistics for an original dataset of 25 values and a subset of 5 values that is removed from it.

	number of values	mean	range
original dataset	25	40	15
removed subset	5	32	8

What can be deduced about the mean and range of the remaining 20 values?

- A mean = 38, range ≤ 15
 B mean = 38, range ≥ 8
 C mean = 38, range ≥ 15
 D mean = 42, range ≤ 15
 E mean = 42, range ≥ 8
 F mean = 42, range ≥ 15

- 5 A combined set of data consists of two groups, P and Q. The number of values in group Q is three times the number of values in group P. The mean of group P is 20 and the mean of the combined set is 32.
What is the mean of group Q?

A 28
B 36
C 44
D 68

- 6 The diagram shows a trapezium $ABCD$ in which AD is parallel to BC , and $\angle DAB = \angle ABC = 90^\circ$.

The lengths of the sides, in centimetres, are:

- $AD = x + 1$
- $AB = 3x - 1$
- $BC = 2x + 4$
- $CD = 3x + 1$

[diagram not to scale]

What is the area, in cm^2 , of the trapezium?

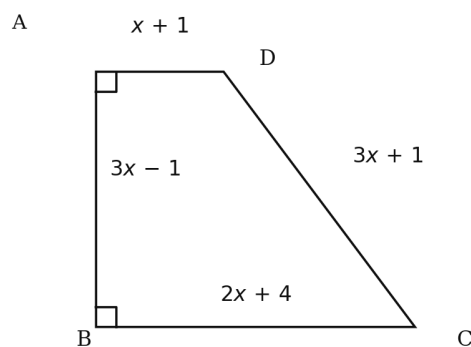


Diagram not to scale

A 24
B 32
C 48
D 56
E 64
F 72
G 112

- 7 The theoretical power available to a wind turbine is directly proportional to the cube of the wind speed.

At a wind speed of v , the turbine operates at an efficiency of 20% and produces an actual power output of 50 kW.

At a wind speed of $2v$, the turbine operates at an efficiency of 15%.

What is the actual power output of the turbine at the wind speed of $2v$?

- A 60 kW
- B 75 kW
- C 150 kW
- D 300 kW
- E 400 kW
- F 1700 kW

- 8 A set of numbers, S_1 , consists of n_1 values, each equal to 10.

A second set of numbers, S_2 , consists of n_2 values, each equal to 30.

When the numbers from S_1 and S_2 are combined, the mean of the resulting set is 14.

A new set is formed by combining the numbers from S_1 with a set of $2n_2$ values, each equal to 30.

What is the mean of this new set?

- A 18
- B $\frac{50}{3}$
- C 22
- D $\frac{130}{7}$
- E $\frac{250}{9}$
- F $\frac{70}{3}$

- 9 The curve with equation $y = ax^2 + bx + c$ has a maximum value of 9 at $x = 2$. The curve passes through the point $(0, 1)$.

What is the value of a ?

- A -4
- B -2
- C -1
- D 1
- E 2
- F 4

- 10 A fair six-sided die with faces numbered 1 to 6 is rolled once.

- If the number rolled is prime, a biased coin with $P(\text{Heads}) = \frac{2}{3}$ is tossed 2 times.
- If the number rolled is not prime, a fair coin is tossed 3 times.

What is the probability that exactly two heads are obtained in total?

- A $\frac{41}{144}$
- B $\frac{43}{108}$
- C $\frac{91}{216}$
- D $\frac{59}{144}$
- E $\frac{7}{16}$
- F $\frac{59}{72}$
- G $\frac{3}{4}$

- 11 A polygon has n sides.

The mean of 5 of its interior angles is 120° .

The mean of its remaining interior angles is 160° .

What is the value of n ?

- A 6
- B 8
- C 9
- D 10
- E 18
- F 48

- 12 The equation

$$\left(\frac{x \times 10^3 + 4x \times 10^2}{7 \times 10^{-1}}\right)^2 - \left(\frac{x \times 10^4 + 2x \times 10^3}{3 \times 10^{-2}}\right) = 2.4 \times 10^7$$

has two real solutions for x .

What is the positive difference between these two solutions?

- A 0.1
- B 2.4
- C 2.5
- D 4.9
- E 9.8
- F 49

- 13 A composite material is formed from two substances, X and Y. Substance X has a density of 3.0 g cm^{-3} and constitutes $\frac{2}{3}$ of the material by volume. Substance Y has a density of 9.0 g cm^{-3} and constitutes the remaining $\frac{1}{3}$ by volume. What is the density of the composite material?

- A 4.0 g cm^{-3}
- B 5.0 g cm^{-3}
- C 6.0 g cm^{-3}
- D 7.0 g cm^{-3}
- E 8.0 g cm^{-3}

- 14** Water flows through a cylindrical pipe of internal diameter 5 cm into a cylindrical tank of internal diameter 2 m.

The depth of the water in the tank increases at a constant rate of 3 cm per minute.

What is the speed of the water in the pipe, in m s^{-1} ?

- A** 0.02
- B** 0.2
- C** 0.8
- D** 3.2
- E** 48
- F** 80

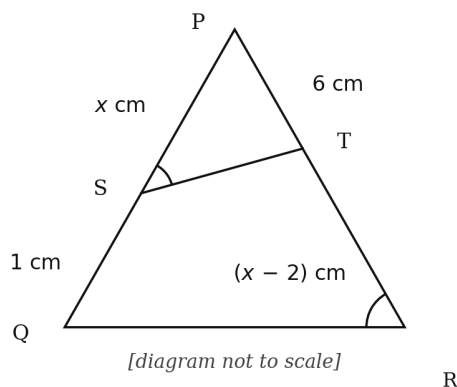
- 15** In the diagram, S is a point on side PQ and T is a point on side PR of triangle PQR .

The angle $\angle PST$ is equal to $\angle PRQ$.

The lengths of the segments are:

- $PS = x$ cm
- $SQ = 1$ cm
- $PT = 6$ cm
- $TR = (x - 2)$ cm

What is the length of side PR ?



- A** 6 cm
- B** 8 cm
- C** 9 cm
- D** 12 cm
- E** $(5 + \sqrt{7})$ cm

- 16** The real numbers x and y satisfy the equation

$$2x^2 - 3xy + 2y^2 = 7$$

What is the maximum possible value of $x - y$?

- A** $\frac{\sqrt{14}}{2}$
- B** 2
- C** $\sqrt{7}$
- D** 4
- E** $2\sqrt{7}$
- F** $4\sqrt{2}$

- 17** A machine extrudes a continuous solid cylinder of plastic of diameter 4 cm. The volume of plastic extruded is 240π cm³ per minute.

What is the speed at which the cylinder emerges from the machine, in mm per second?

- A** 1
- B** 2.5
- C** 10
- D** 40
- E** 60
- F** 600

- 18** Seven distinct numbers are written in strictly increasing order:

$$x_1 < x_2 < x_3 < x_4 < x_5 < x_6 < x_7$$

Their mean is M , their median is m , and their range is R .

The smallest and largest of these seven numbers are then removed, leaving five numbers.

Which of the following statements must be true?

- 1 The median of the remaining five numbers is equal to m .
- 2 The range of the remaining five numbers is strictly less than R .
- 3 If $M = m$, the mean of the remaining five numbers is equal to M .

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 19** The diagram shows a right-angled trapezium with parallel horizontal sides of lengths $2x$ and $3x$, and vertical height h . The area of this trapezium is A_1 .

A new trapezium is formed by modifying these dimensions:

- the side of length $3x$ is increased by 40%
- the side of length $2x$ is decreased by 35%
- the vertical height h is increased by 20%

The area of the new trapezium is A_2 .

The relationship between A_1 and A_2 can be expressed as $A_2 = kA_1$.

What is the value of k ?

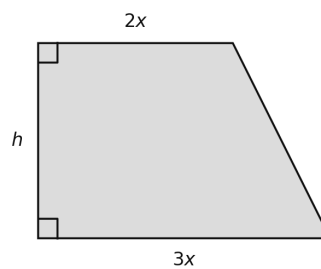


Diagram not to scale

- A** $\frac{27}{25}$
- B** $\frac{57}{50}$
- C** $\frac{5}{4}$
- D** $\frac{33}{25}$
- E** $\frac{7}{5}$

- 20** The tensile strengths of a batch of 50 manufactured fibres are tested. The batch consists of 30 carbon fibres and 20 glass fibres.

Statistics for the carbon fibres and for the entire batch of 50 fibres are shown in the table:

Fibre group	Number of fibres	Mean tensile strength / GPa	Range of tensile strength / GPa
Carbon fibres	30	3.6	1.1
All 50 fibres combined	50	3.2	1.6

What can be deduced about the mean and range of the tensile strength of the 20 glass fibres?

- A** mean = 2.6 GPa, range $\leq$ 1.6 GPa
B mean = 2.6 GPa, $1.1 \text{ GPa} \leq \text{range} \leq 1.6 \text{ GPa}$
C mean = 2.6 GPa, range $\geq$ 1.6 GPa
D mean = 2.8 GPa, range $\leq$ 1.6 GPa
E mean = 2.8 GPa, $1.1 \text{ GPa} \leq \text{range} \leq 1.6 \text{ GPa}$
F mean = 2.8 GPa, range $\geq$ 1.6 GPa

- 21** The region R in the xy -plane is defined by the inequalities:

$$x \geq 0, \quad y \geq 0, \quad x + y \leq 6, \quad 3x + y \leq 12$$

What is the area of R ?

- A** 3
B 9
C 12
D 15
E 18

- 22** The following six numbers are given in ascending order:

$$10, 16, h, 2h, 38, 44$$

The mean of these six numbers is equal to their median.

What is the value of h ?

- A** 7.2
- B** 12.0
- C** 13.5
- D** 18.0
- E** 24.0
- F** 36.0

- 23** It takes time T for a single pipe to fill a particular cylindrical tank.

A new cylindrical tank is used, which has three times the radius and one-third the height of the original tank.

This new tank is filled using 6 pipes simultaneously. Each of these pipes has a radius that is twice the radius of the original pipe, and the water flows through each at half the speed of the water in the original pipe.

In terms of T , what is the time taken to fill the new tank?

- A** $\frac{1}{12}T$
- B** $\frac{1}{8}T$
- C** $\frac{1}{4}T$
- D** $\frac{1}{2}T$
- E** $\frac{3}{4}T$

- 24** Find the set of values of the real constant k for which the line with equation $y = 2x$ does not intersect the circle with equation $x^2 + y^2 - 4x + 6y + k = 0$.

- A** $16/5 < k < 13$
- B** $k > 16/5$
- C** $k < 13$
- D** $k < 16/5$
- E** $k > 13$
- F** $16/5 \leq k < 13$

- 25** A square has side length $2a$.

A circle, C , is inscribed in the square.

A quarter-circle, Q , has its centre at one vertex of the square and has a radius equal to the side length of the square.

What is the area of Q minus the area of C ?

- A** 0
- B** $2\pi a^2$
- C** $3\pi a^2$
- D** $-3\pi a^2$
- E** $a^2(4 - \pi)$
- F** πa^2

- 26** Given that $p = 2^x$ and $q = 3^x$, which one of the following is an expression for

$$\frac{6^{x+1} \times 12^{x-1}}{18^{2x-1}}$$

in terms of p and q ?

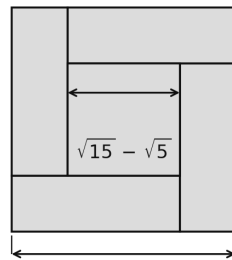
- A** $\frac{p}{q^2}$
- B** $\frac{4p}{q^2}$
- C** $\frac{9p}{q^2}$
- D** $\frac{9p}{q}$
- E** $\frac{p}{9q^2}$
- F** $9pq^2$

- 27** The diagram shows four identical shaded rectangles arranged inside a large square to leave a central unshaded square.

The side length of the outer square is $\sqrt{15} + \sqrt{5}$.

The side length of the inner unshaded square is $\sqrt{15} - \sqrt{5}$.

What is the total area of the shaded region?



$$\sqrt{15} + \sqrt{5}$$

Diagram not to scale

- A** 0
- B** $5\sqrt{3}$
- C** $10\sqrt{3}$
- D** $20\sqrt{3}$
- E** 40
- F** $40 + 20\sqrt{3}$
- G** $40\sqrt{3}$